

Table of Contents

1 XML Digital Signature	3
1.1 Procedure for message digital signing and message digital signature verification:	4
1.2 Sample of digitally signed message	4

0.1 XML Digital Signature

Participants use certificates issued by IPS for message digital signing. To make it possible for other trusted parties verify a digital signature, the public part of the certificate must be incorporated in the attributes of the digital signature by party who sign the message. Message digital signature schema implements W3C xmldsig-core schema. The enveloped digital signature enclosing method described in the W3C xmldsig-core schema should be used for the purpose of attaching a digital signature. This means that the digital signature will not be included in the message checksum calculation. Enveloped signature method should be used as a method of attaching a digital signature to message.

ELEMENT	ATTRIBUTES	DESCRIPTION
Signature		Root element of an XML Signature.
SignedInfo		SignedInfo includes the canonicalization algorithm, a signature algorithm, and one or more references.
CanonicalizationMethod	Algorithm="http://www.w3.org/TR/2001/REC-xml-c14n-20010315"	CanonicalizationMethod is a required element that specifies the canonicalization algorithm applied to the SignedInfo element prior to performing signature calculations.
SignatureMethod	Algorithm="http://www.w3.org/2001/04/xmldsig-more#rsa-sha256"	Algorithm of the digital signature.
Reference	URI=""	Defines the reference to the message section to be digitally signed; as the entire message will be digitally signed and the digital signature will be added after signing, URI="" has to be indicated.
Transforms		Transforms element contains an ordered list of Transform elements; these describe how the signer obtained the data object that was digested.
Transform	Algorithm="http://www.w3.org/2000/09/xmldsig#enveloped-signature"	Method used for adding the digital signature.
Transform	Algorithm="http://www.w3.org/TR/2001/REC-xml-c14n-20010315"	Method used for adding the digital signature.
DigestMethod	Algorithm="http://www.w3.org/2001/04/xmlenc#sha256"	Checksum calculation algorithm.
DigestValue		Message checksum. The value has to be indicated in base64 encoding.

ELEMENT	ATTRIBUTES	DESCRIPTION
SignatureValue		Digital signature of the message. The value has to be indicated in base64 encoding.
KeyInfo		KeyInfo is an element that enables the recipient to obtain the key needed to validate the signature.
X509Data		An X509Data element within KeyInfo contains one X509 certificate.
X509Certificate		Certificate used for digitally signing the message. The value has to be indicated in base64 encoding.

Note: Use of KeyValue is discouraged, as it introduces an ambiguity and a security hazard. The public key used to sign the document is already encoded in the certificate (which is in X509Data).

0.0.1 Procedure for message digital signing and message digital signature verification:

1. The debtor's agent digitally signs message using his own private key and certificate signed by IPS for this private key and sends this message to the IPS;
2. IPS establish trust in digital signature certificate against his trusted certificate chain and then verifies the digital signature of the message;
3. IPS digitally signs the message using his own private key and public certificate for this private key and sends the message to the creditor's agent;
4. The creditor's agent establish trust in digital signature certificate against certificates chain published by IPSL and verifies the digital signature of the message.

0.0.2 Sample of digitally signed message

```
<?xml version="1.0" encoding="UTF-8"?>
<Document xmlns="urn:iso:std:iso:20022:tech:xsd:acmt.023.001.02">
  <IdVrfctnReq>
    <Assgnmt>
      <MsgId>3003cb156d294c869b937c37d3c210cc</MsgId>
      <CreDtTm>2021-02-19T17:37:59Z</CreDtTm>
      <FrstAgt>
        <FinInstnId>
          <Othr>
            <Id>0020</Id>
          </Othr>
        </FinInstnId>
      </FrstAgt>
      <Assgnr>
        <Agt>
          <FinInstnId>
            <Othr>
              <Id>1000</Id>
            </Othr>
          </FinInstnId>
        </Agt>
      </Assgnr>
      <Assgne>
        <Agt>
          <FinInstnId>
            <Othr>
```

